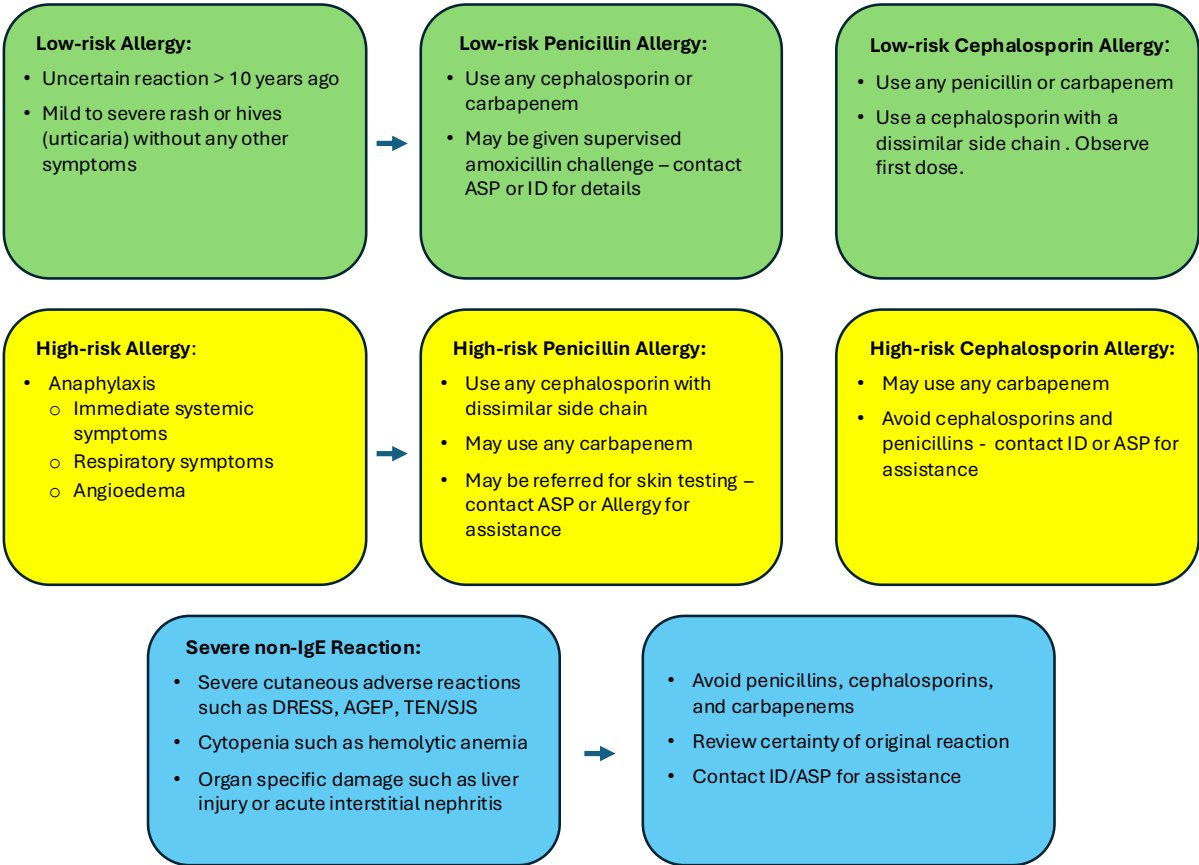


Beta-Lactam Reactions
Assessment of Reaction & Selection of Alternative Beta-Lactams

- Penicillin allergies wane over time
- Cephalosporins are preferred over non-beta-lactams for safety and effectiveness.
- Beta lactam cross-reactivity is driven by similar side chains.
- **Patients with a history of anaphylactic reaction to penicillin can receive a cephalosporin with non-related side chains, such as cefazolin, without further testing or precaution.**
- Use **Figure 1** for assessment of allergies. See **Figure 2**, page 2, for shared side chains.

Figure 1. Assessment of Beta-lactam Allergy



Beta-lactam Antibiotic Cross-reactivity Chart

- Cross-reactivity charts should **only** be consulted after assessing severity of reaction.
- Consult ID or ASP for assistance in assessing specific cases

Figure 2. Beta-lactam Antibiotic Cross-reactivity Chart

	"X" Caution - Contact ID/ASP for guidance	Penicillin	Ampicillin	Amoxicillin	Cefazolin	Cephalexin	Cefadroxil	Cefoxitin	Cefaclor	Cefprozil	Cefuroxime	Cefotaxime	Ceftriaxone	Cefdinir	Cefpodoxime	Ceftazidime	Cefixime	Cefepime	Cefiderocol	Aztreonam
Penicillins	Penicillin		X	X		X	X		X	X										
	Ampicillin	X		X		X	X		X	X										
	Amoxicillin	X	X			X	X		X	X										
1st Gen	Cefazolin																			
	Cephalexin	X	X	X			X		X	X										
	Cefadroxil	X	X	X		X			X	X										
2nd Gen	Cefoxitin										X									
	Cefaclor	X	X	X		X	X			X										
	Cefprozil	X	X	X		X	X		X											
	Cefuroxime							X												
3rd Gen	Cefotaxime												X		X			X		
	Ceftriaxone											X		X				X		
	Cefdinir															X				
	Cefpodoxime											X	X					X		
	Ceftazidime																		X	X
	Cefixime												X							
Later Gen	Cefepime											X	X		X					
	Cefiderocol														X					X
Mono-bactam	Aztreonam														X				X	

"X" – Caution – Contact ID/ASP for guidance. Ref: Khan et al 2022, Reddy et al 2026, adsp.nm.org

References

- [Khan et al, Drug allergy: A 2022 practice parameter update J Allergy Clin Immunol, 2022, 150\(6\)133](#)
- [Chaudhry et al., Cephalosporins: A Focus on Side Chains and ̢-Lactam Cross-Reactivity. Pharmacy, 2019, 7\(3\)103](#)
- [Shenoy et al., Evaluation and Management of Penicillin Allergy. JAMA. 2019, 321\(2\)188](#)
- [Reddy et al, Intracameral Cefuroxime Use in Cataract Surgery After Penicillin Allergy Reclassification JAMA Ophthalmol. 2026, 2026, 144\(4\):325](#)